Microsoft

The Way of Cloud Native AI

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Hello!

From our family ☺

CLOUD NATIVE
COMMUNITY DAYS
AMSTERDAM

Agenda

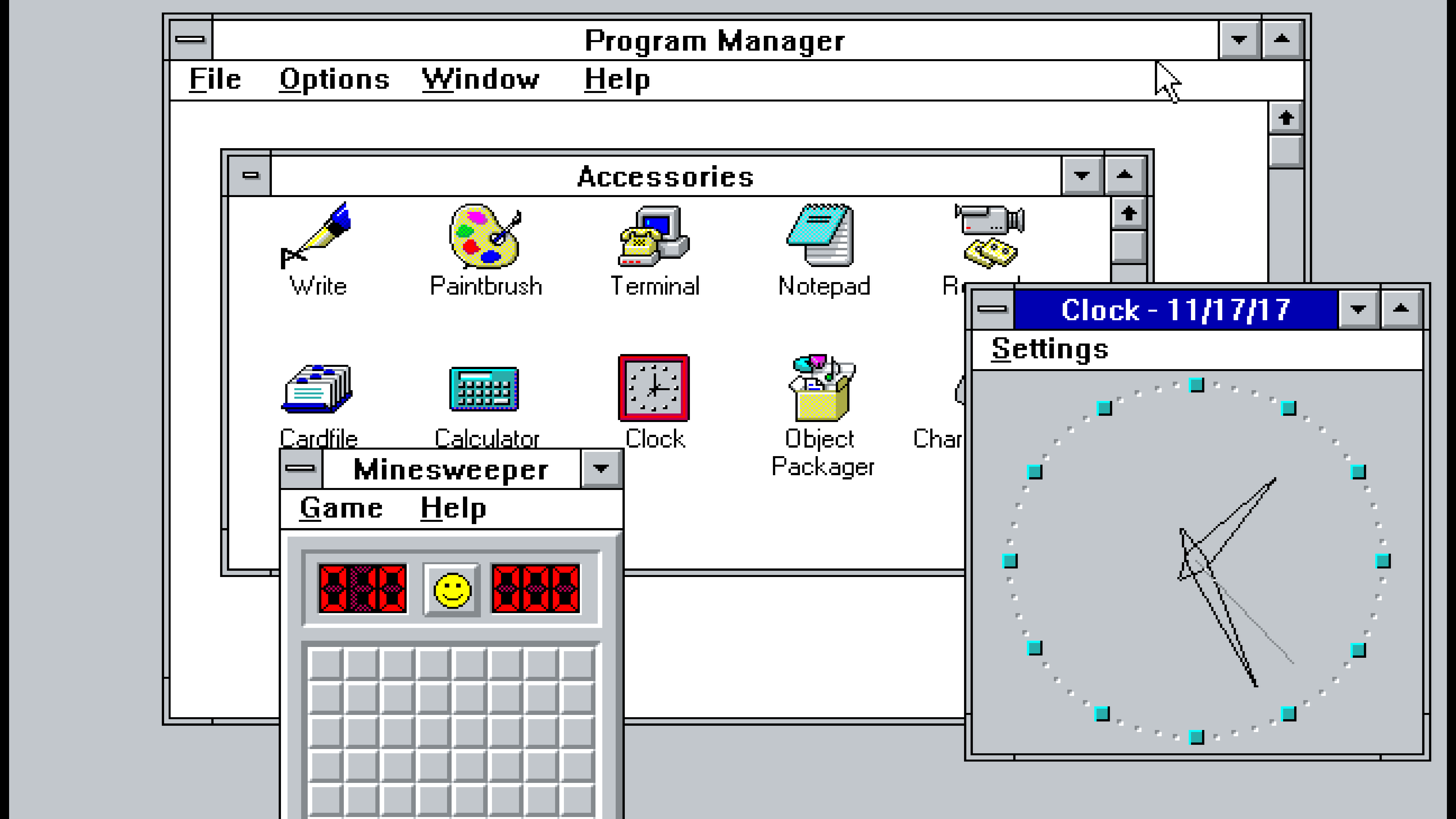
1. A personal story
2. Bringing AI capabilities to Cloud Native workloads
3. Infusing AI into existing apps
4. Using AI to support Cloud Native infrastructure

A personal computing journey

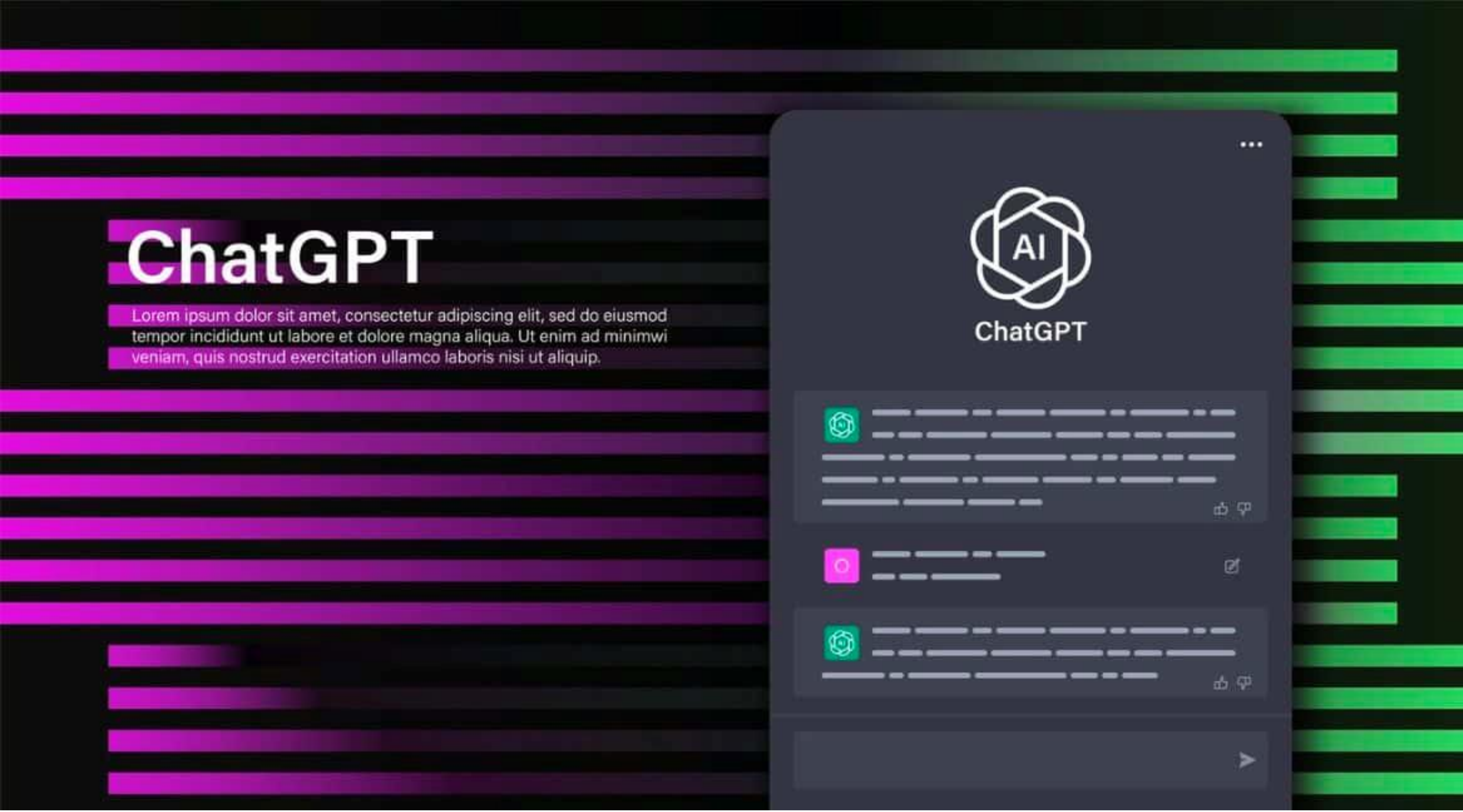

commodore 64
POWER


AmigaOS 4.1 Update 2 running on AMCC PPC460EX with 2,048MB Memory and Radeon RV730 not yet implemented
Utilities
System
21:05:28
Mo Tu We Th Fr Sa Su
AmiUpdate
AmiVNC4
Devs
Documentation
Internet
Mount SMB Drives
MUI
Prefs
Samba
Storage
System
Utilities
AmiDVD
AmiGS
AmiPDF
Blankers
Commodities
DiskMaster2
Dockies
DOpus4
DvPlayer
Filer
Ghostscript
Ranger
SGrab
Tunenet
MakeCD
Clock
GraphicDump
IconEdit
IntPrinter
Installation Utility
KeyShow
MEmacs
Multiview
NotePad
PartitionWizard
PlayCD
PrefsObjectsEditor
PrintFiles
RawDisk
ShowConfig
UnArc
USBInspector
Calculator
Python
Find
FixFonts
Format
FormatCDRW
DiskImageGUI
Help
Media Toolbox
Mounter
RexxMast
RinghioServer
TypeManager
Shell
Prefs
Presets
AHI
AmigaInput
ASL
Compatibility
DefIcons
DOS
Font
GUI
Input
Internet
Locale
Notifications
Palette
Picasso96Mode
Pointer
PopupMenu
Printer
PrinterPS
PrinterGfx
ScreenBlanker
ScreenMode
Screens
Serial
Sound
Time
Timezone
URL
USB
WBPattern
WBStartup
Workbench
updates
Env-Archive
Internet
Action logs
AmiFTP
Connection logs
Connections
Edit Connection
IBrowse
New Connection
Open Connection
OWB
pFTP
RDesktop
Reports
SimpleMail
Templates
YAM
Mac OS X Music
RAM Disk
OS4.1
WORK

```
220-------------------------------------------------------------------------------
220-                       R S Y N C . O S U O S L . O R G
220-                           Oregon State University
220-                               Open Source Lab
220-
220-         Unauthorized use is prohibited - violators will be prosecuted
220-------------------------------------------------------------------------------
220-
220-                  For more information about the OSL visit:
220-                      http://osuosl.org/services/hosting
220-
220-            This host is the home to the primary archives of several
220-             projects.  We would prefer that only primary/secondary
220-                      mirrors use this service.  Thanks!
220-
220-------------------------------------------------------------------------------
220-
220
Name (ftp.slackware.com:klaatu): anonymous
Password (ftp.slackware.com:anonymous):
331 Please specify the password.
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```


Program Manager
File
Options
Window
Help
Accessories
Write
Paintbrush
Terminal
Notepad
Cardfile
Calculator
Clock
Object Packager
Minesweeper
Game
Help
Clock - 11/17/17
Settings

12:00
NEVER
SETTLE
October 4
October 4


ChatGPT
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minimwi veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip.
AI
ChatGPT

Blog

May 13, 2024

Introducing GPT-4o and more tools to ChatGPT free users

We are launching our newest flagship model and making more capabilities available for free in ChatGPT.

Try ChatGPT ↗ Rewatch the livestream >

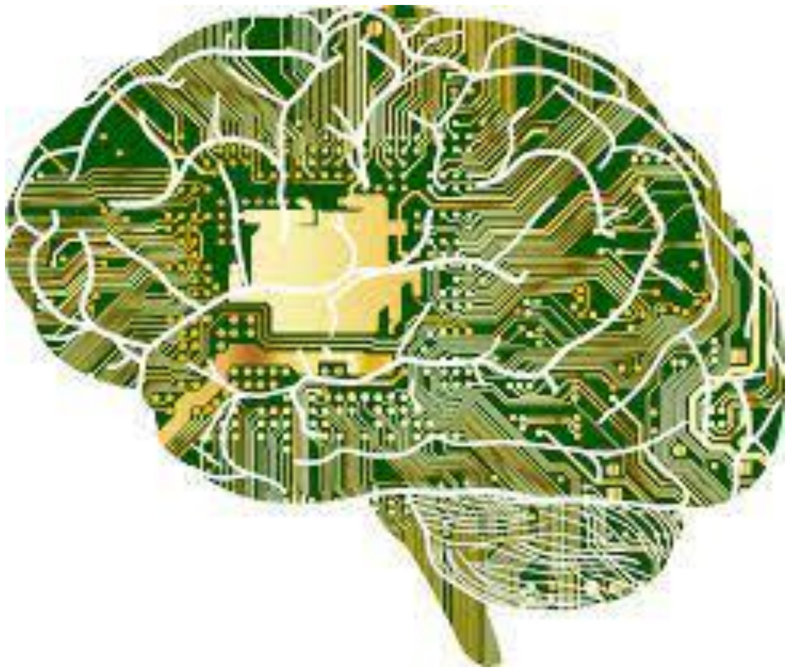
November 30, 2022

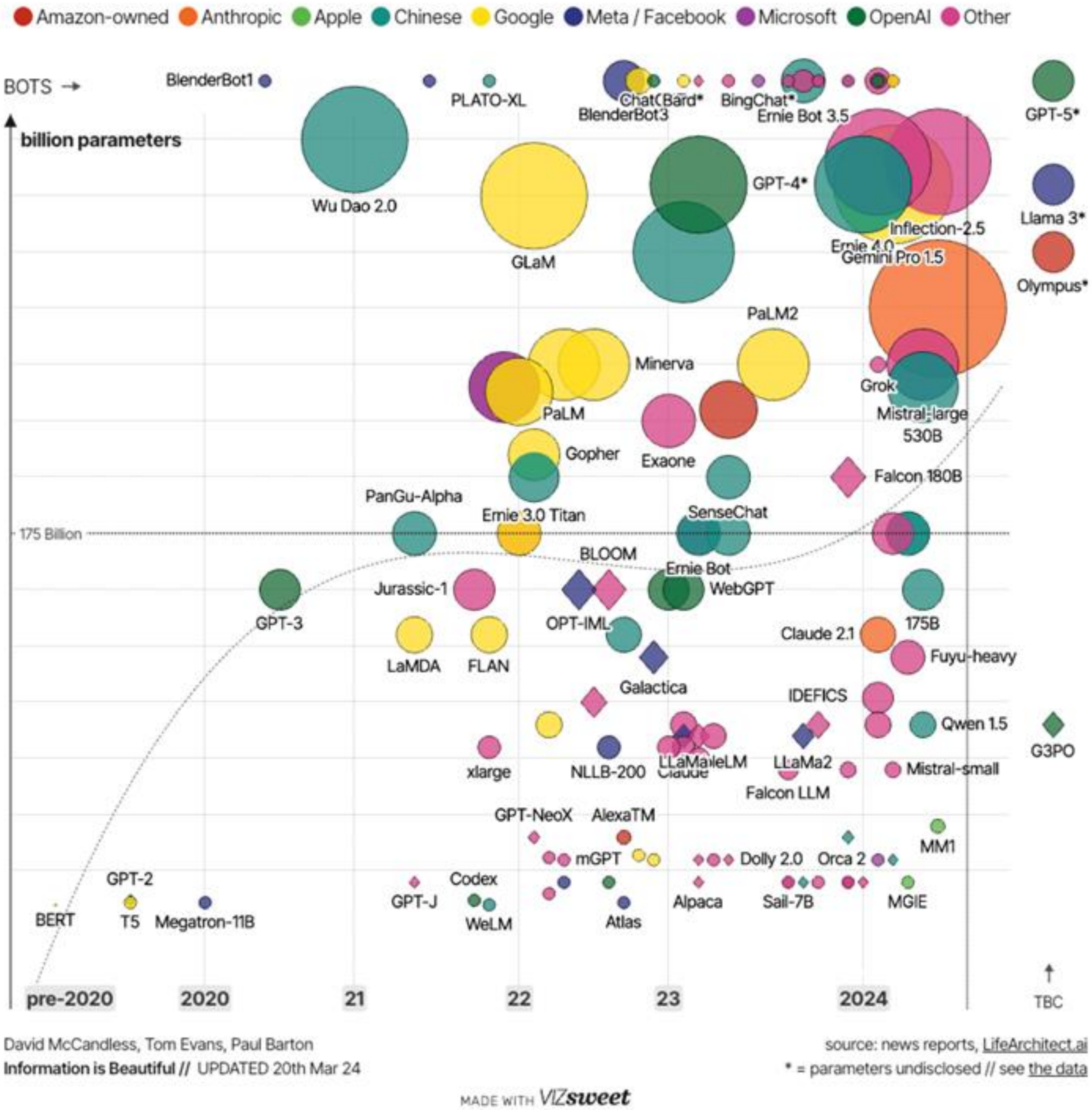
Authors

OpenAI ↓


LLM! LLM!
LIGHTYEAR
EVERYWHERE
imgflip.com

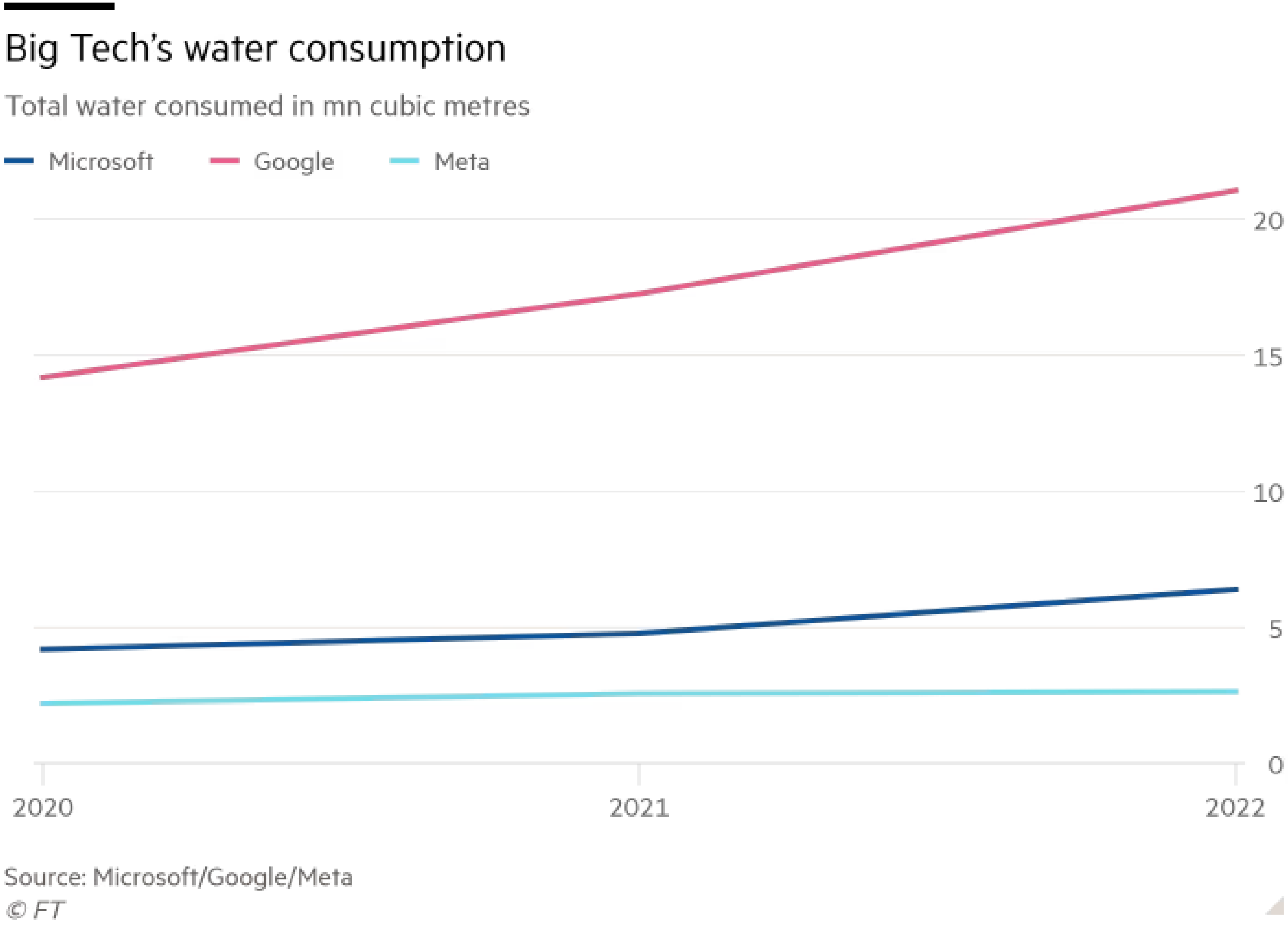
The Rise of LLMs





The cost of AI

"A month before OpenAI finished training its most advanced model, GPT-4, a data centre cluster in West Des Moines, Iowa, consumed 6 per cent of the district's water, according to a lawsuit filed by its residents."



Intelligent Apps

Generative AI makes apps truly intelligent

Current apps

- Constrained interactions
- Hard-coded and fixed data sets
- Change is costly and complex

Paradigm Shift

Intelligent apps

- Natural language interaction
- Data-driven, personalized experiences
- Continuously learns and improves

Using OpenAI models

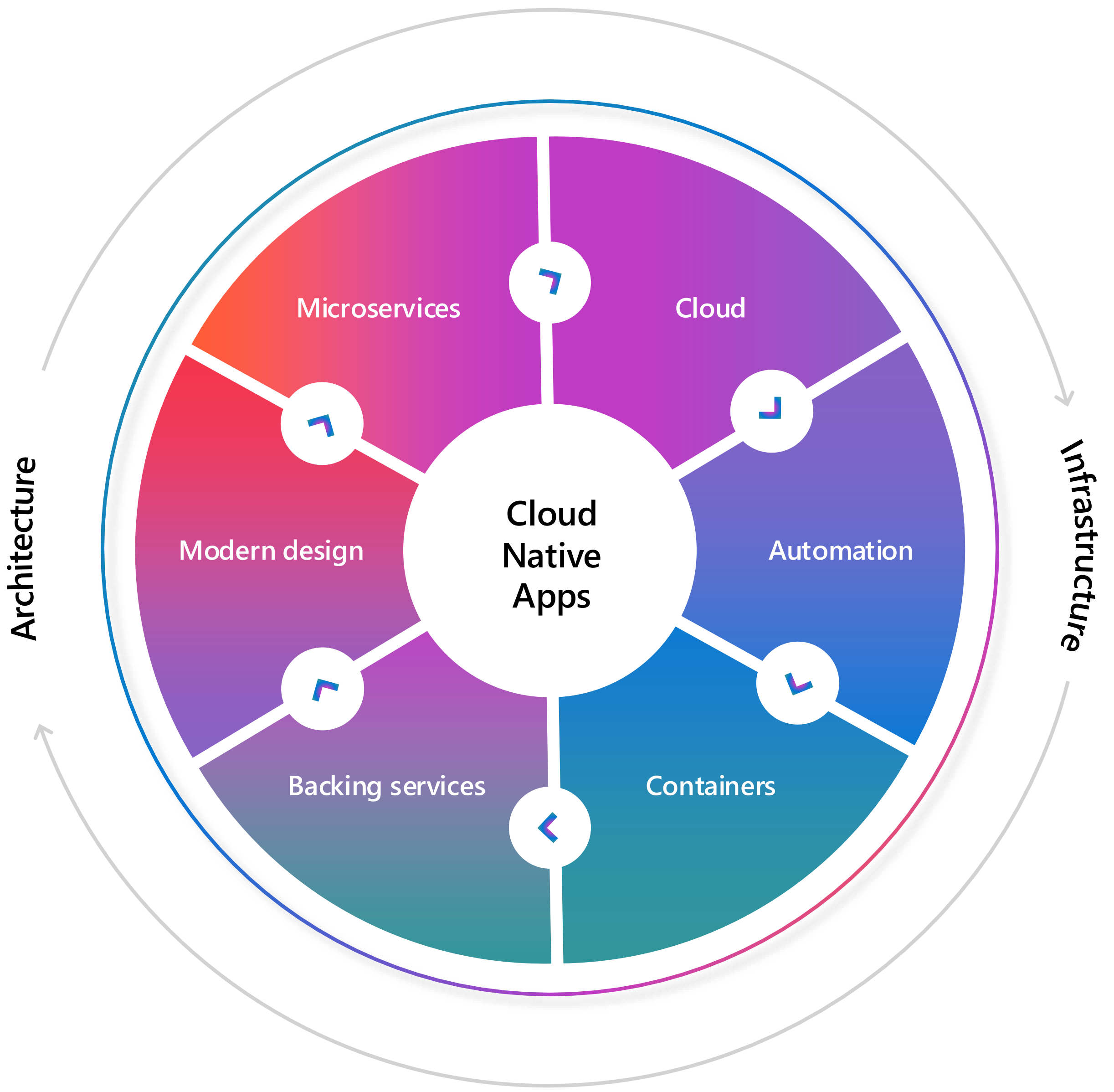
```
curl https://api.openai.com/v1/chat/completions \
  -H "Content-Type: application/json" \
  -H "Authorization: Bearer $OPENAI_API_KEY" \
  -d '{
     "model": "gpt-3.5-turbo",
     "messages": [{"role": "user", "content": "Say this is a
test!"}],
     "temperature": 0.7
   }'
```

Using Azure OpenAI models

```
curl
https://myaccount.openai.azure.com/openai/deployments/mymodel/chat/completions?api-version=2023-05-15 \
  -H "Content-Type: application/json" \
  -H "api-key: YOUR_API_KEY" \
  -d '{"messages":[{"role": "system", "content": "You are a helpful assistant."},{"role": "user", "content": "Does Azure OpenAI support customer managed keys?"},{"role": "assistant", "content": "Yes, customer managed keys are supported by Azure OpenAI."},{"role": "user", "content": "Do other Azure AI services support this too?"}]}'
```

Cloud Native App Considerations

Cloud Native Capabilities

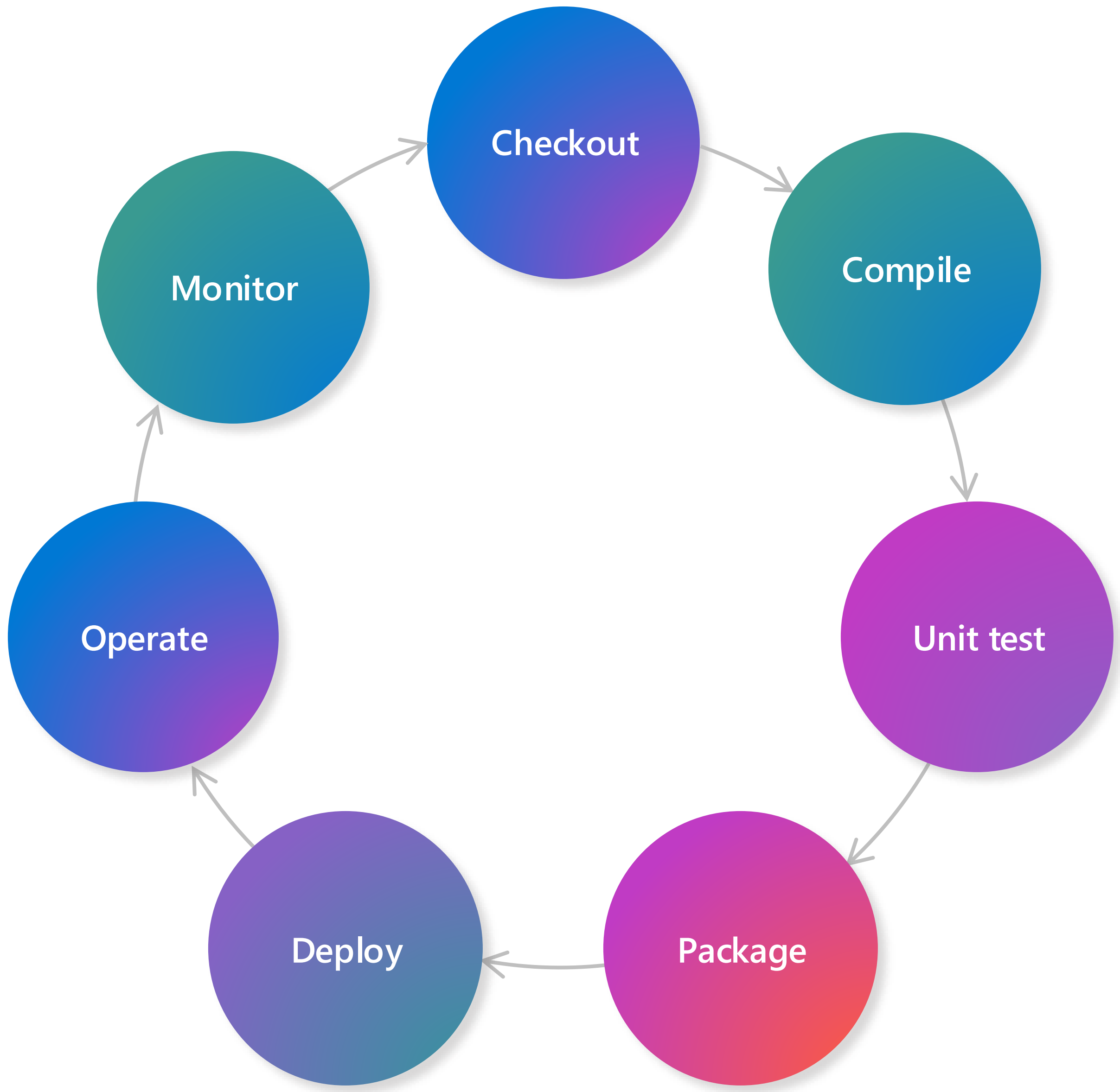


Live in the cloud

Modern design

- Treat everything as a loosely-coupled service, decoupled from infrastructure components
- Built with best of breed technologies
- Communicate with lightweight APIs and protocols, such as REST and gRPC
- Stateless and massively scalable
- Containerized and orchestrated
- Embrace DevOps and automation

Automate the things



Things to consider when building with AI…

Microsoft AI principles:

- Fairness
- Reliability and safety
- Privacy and security
- Inclusiveness
- Accountability
- Transparency

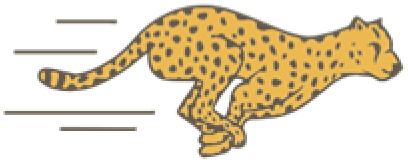
Prompting with frameworks and/or SDKs

Deploy with resiliency and scale in mind

Cloud Native + AI

What is CNAI

CNCF TAG Runtime



https://tag-runtime.cncf.io/

OPENAI

Scaling Kubernetes to 7,500 nodes

HUGGING FACE

Hugging Face Collaborates with Microsoft to launch Hugging Face Model Catalog on Azure

Cloud Native Artificial Intelligence Whitepaper

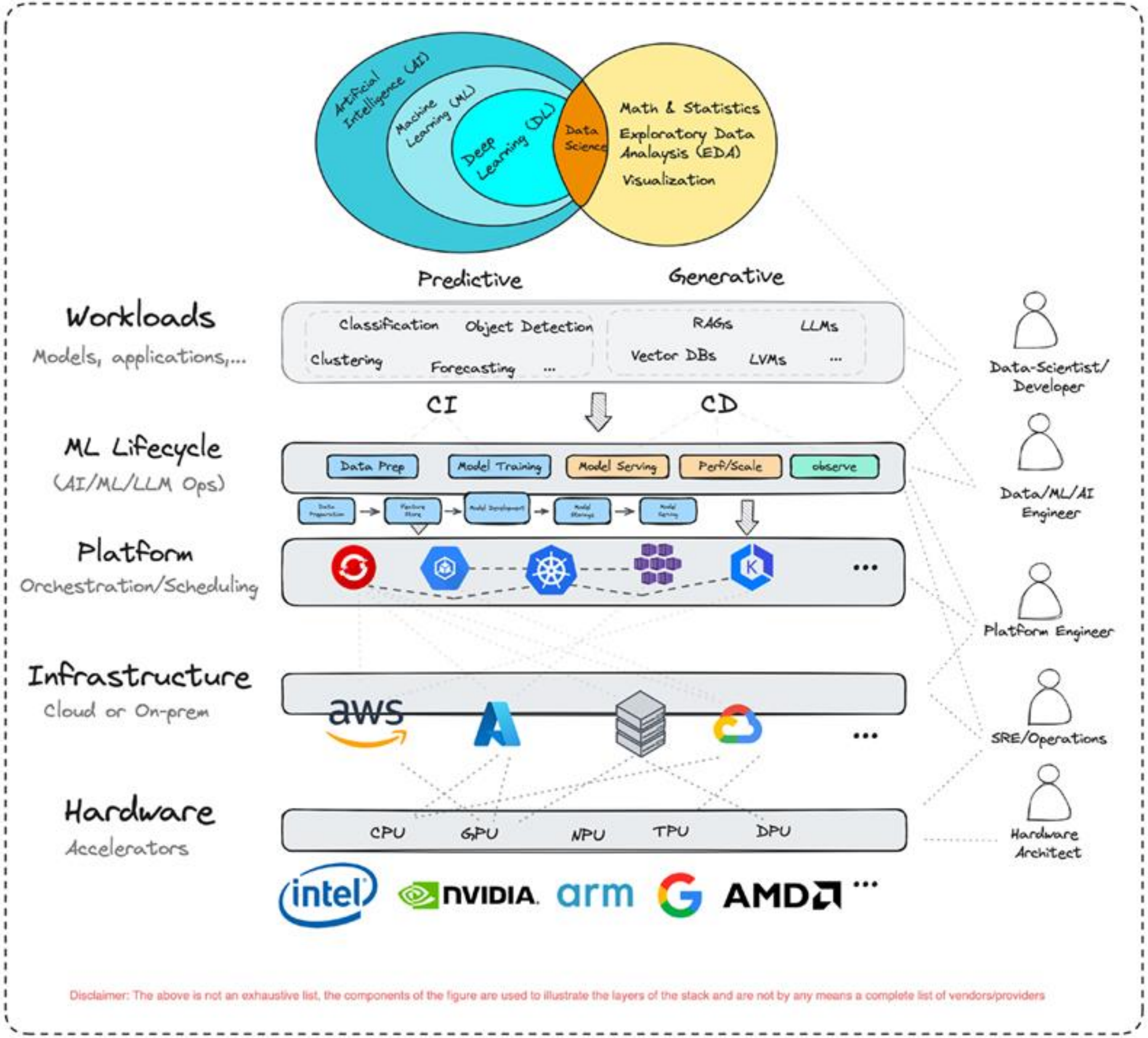
PUBLISHED: MARCH 19, 2024

Cloud Native and Artificial Intelligence are the most critical technology trends today. Cloud Native technology provides a scalable and reliable platform for running applications. Given recent advances in AI and Machine Learning, it is steadily rising as a dominant cloud workload. While Cloud Native technologies readily support certain aspects of AI/ML workloads, challenges and gaps remain, presenting opportunities to innovate and better accommodate.

This paper presents a brief overview of the state-of-the-art AI/ML

https://www.cncf.io/wp-content/uploads/2024/03/cloud_native_ai24_031424a-2.pdf

What is CNAI


Artificial Intelligence (AI)
Machine Learning (ML)
Deep Learning (DL)
Data Science
Math & Statistics
Exploratory Data Analaysis (EDA)
Visualization
Predictive
Generative
Workloads
Models, applications,...
Classification
Object Detection
Clustering
Forecasting
...
RAGs
LLMs
Vector DBs
LVMs
...
CI
CD
ML Lifecycle
(AI/ML/LLM Ops)
Data Prep
Model Training
Model Serving
Perf/Scale
observe
Platform
Orchestration/Scheduling
...
Infrastructure
Cloud or On-prem
...
Hardware
Accelerators
CPU
GPU
NPU
TPU
DPU
...
Data-Scientist/ Developer
Data/ML/AI Engineer
Platform Engineer
SRE/Operations
Hardware Architect
Disclaimer: The above is not an exhaustive list, the components of the figure are used to illustrate the layers of the stack and are not by any means a complete list of vendors/providers

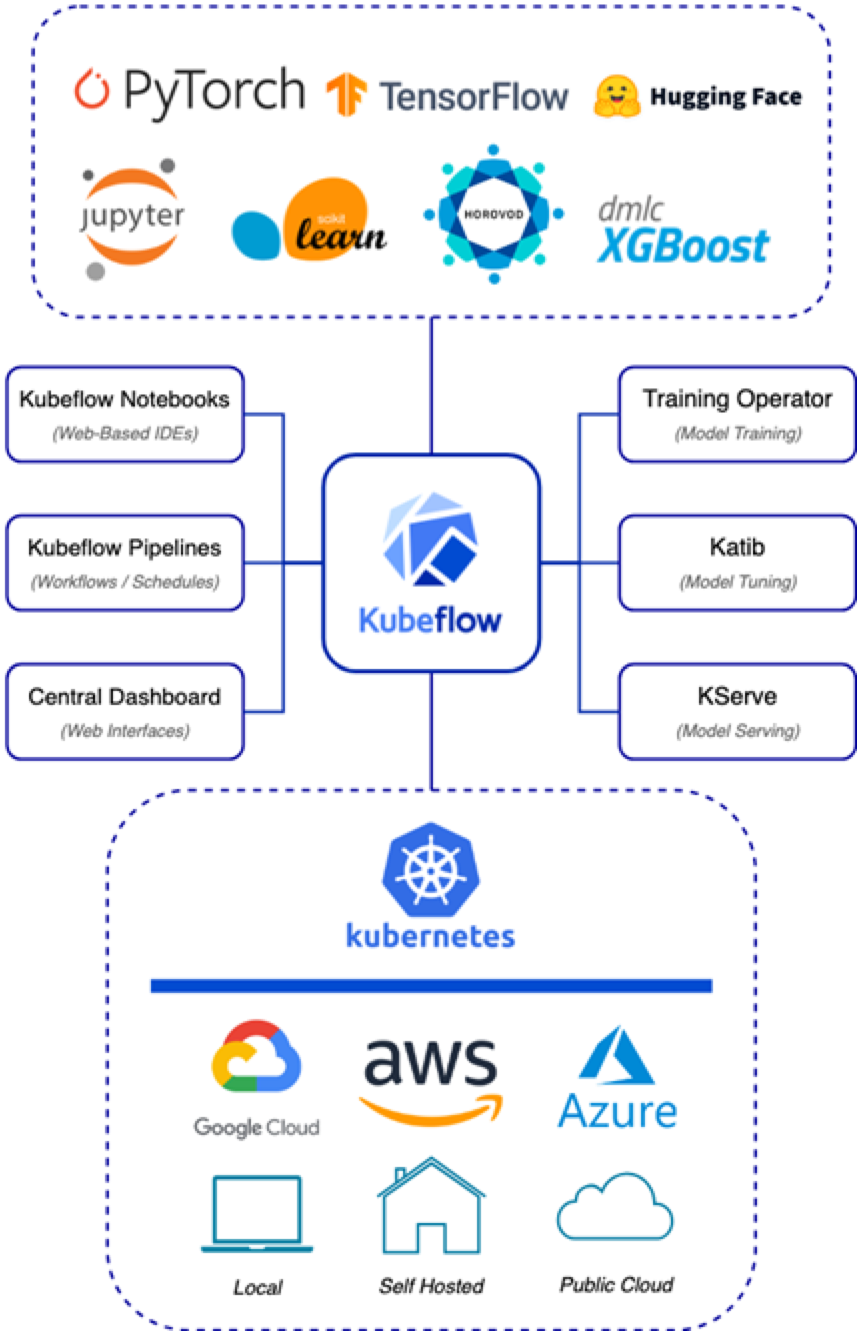
CNCF AI Projects

Kubeflow

VOLCANO

K8SGPT
KUBERNETES
SUPERPOWERS

LLM•NETES



Azure Kubernetes Service (AKS)

Enterprise grade by design

- Built-in best practices
- Multi-layer security
- Increased operations efficiency

First class E2E portfolio of solutions for your apps
Full range of integrated and managed solutions for your apps such as Azure Monitor managed Prometheus and Grafana, Azure Backup for AKS complemented via ASO by top performant databases, messaging, storage

Globally scalable and reliable services
With 99.95% SLA, scaling to 5k nodes, presence in all Azure regions and clouds + edge, integration with Azure Load Testing and Azure Chaos Studio

Unified Dev and Ops of at-scale cloud and hybrid environments
Kubernetes Fleet Manager allows you to command your workloads, development and operations of all your Azure Kubernetes clusters, whether they are AKS, Hybrid and Edge AKS or any K8s cluster onboarded via Arc

Rich ecosystem of integrated partners and solutions
Self-serve marketplace for Kubernetes and cloud native application that allows you to do one-click deployment of fit-for-purpose business solutions from our amazing partner ecosystem

Unrivalled performance
Mold the platform to maximize your workload's performance while remaining fully managed. Fastest and most flexible network options

End-to-end Cloud Native path

Community

Security

Microsoft Entra ID

Azure Policy

Microsoft Defender

AI

AI Services

OpenAI

Machine Learning

Development tools

GitHub

Visual Studio Code

Azure API Management

Azure Container Registry

Platform

Azure Kubernetes Service

Azure Kubernetes Fleet Manager

Managed databases

Azure PostgreSQL

Azure Cosmos DB

Azure SQL family

Azure MySQL & MariaDB

Storage

Azure Storage

Cloud operations anywhere

Azure Arc

Introducting KAITO

To run open-source LLMs on k8s, you'll need to:

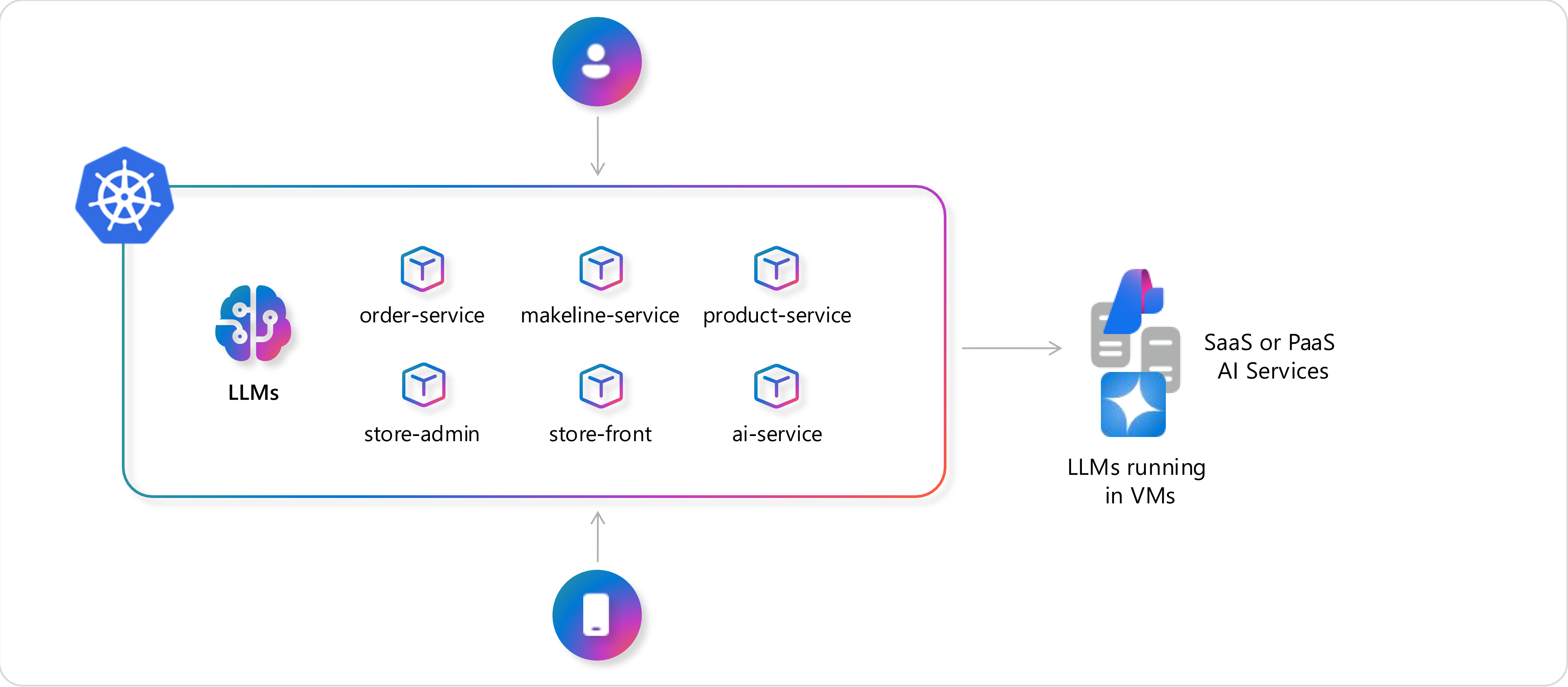
1. Wait for model weights to load
2. Reduce weights into a container image
3. Host the model containers in a private registry
4. Provision right-sized, compatible GPU infrastructure
5. Tune model deployment parameters to the hardware
6. Troubleshoot shared GPU quota limitations
7. Troubleshoot different GPU hardware
8. Set up inference server as an app endpoint service

…

Custom Open Source LLM's – Why?

Avoid vendor lock-in	More portable across cloud (similar to the managed Kubernetes)
Control and Customization	Ability to choose any model and customize/re-train as needed
Accuracy	Can fine tune to meet unique requirements
Transparency	Assists with audits and ethical/legal compliance
Security	Keep models and data isolated
Data residency	Country based (particularly FSI)
Cost	No licensing fees/premiums*, flexibility in hardware * Read the fine print

Leveraging LLMs with your applications

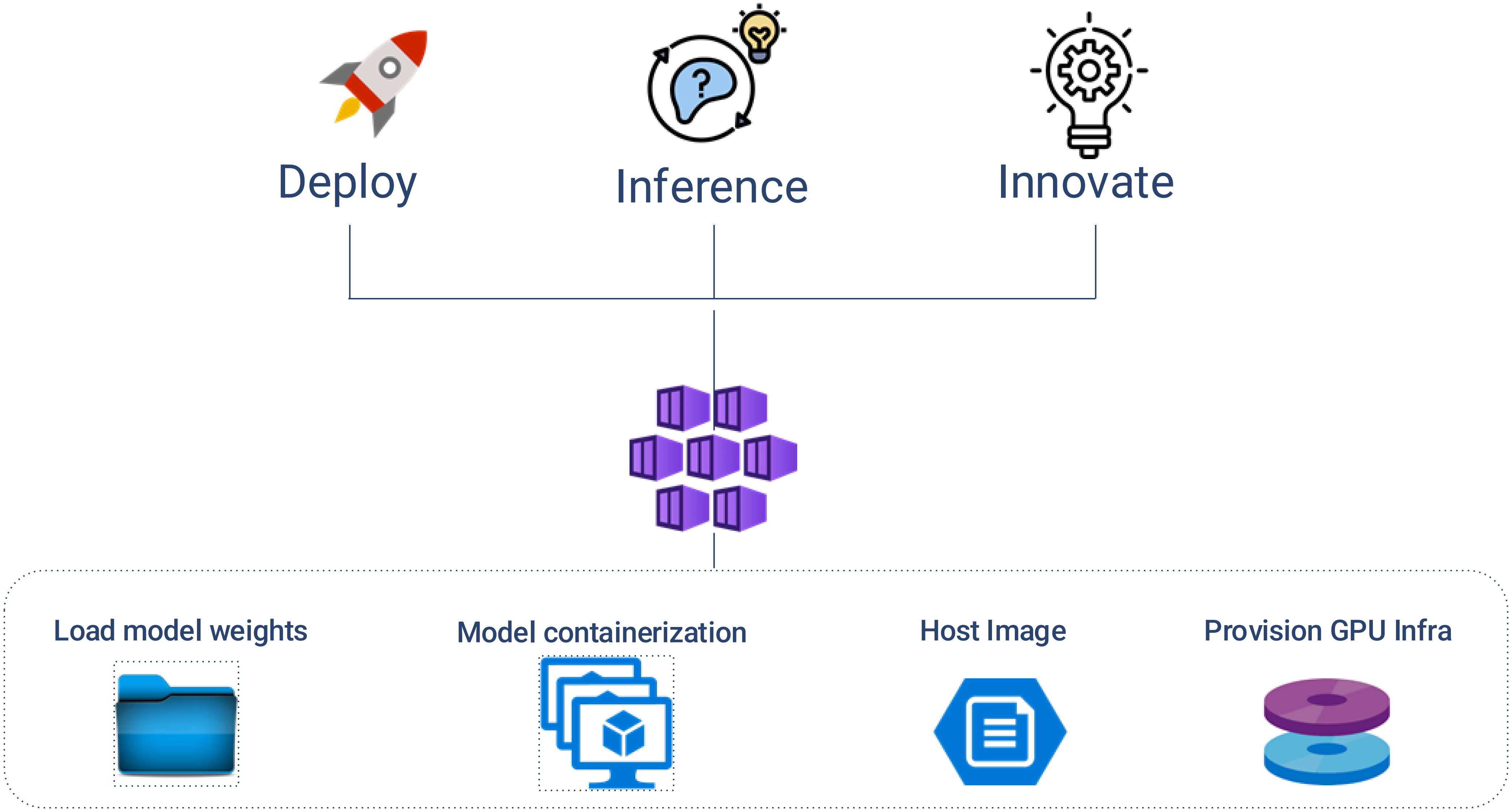


Kubernetes for AI

Kubernetes	AI	Toolchain	Operator
Leverage the ecosystem		Deployment tools other than a "Platform/Service"	Streamline workflows

thenewstack.io/jumpstart-ai-workflows-with-kubernetes-ai-toolchain-operator/

Introducing Kaito



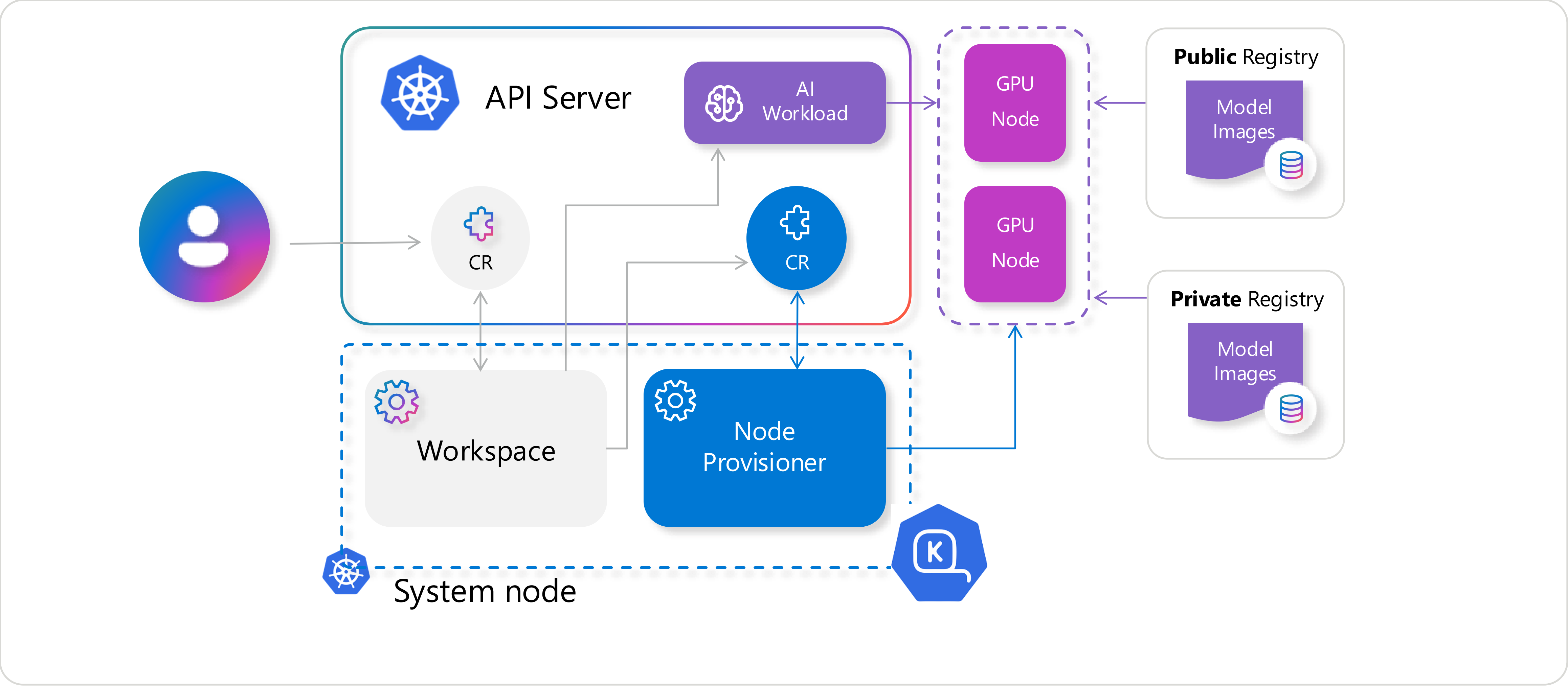
Kaito - Kubernetes AI Toolchain Operator

Targeted towards deploying large open source language models, such as falcon and llama2

- Manage large model files using container images, a http server is provided to perform inference calls using the model library.
- Avoid tuning deployment parameters to fit GPU hardware by providing preset configurations.
- Auto-provision GPU nodes based on model requirements.
- Host large model images in the public registry - if the license allows.

Using Kaito, the workflow of onboarding large AI inference models in k8 is simplified.

KAITO architecture



Kaito: Reduce time to inference

Current
1. Environment setup (GPU nodes, drives, plugins_ 2. Model setup 3. Clone repo 4. Model weights 5. Containerization 6. Model dependencies 7. Inference code 8. Image storage 9. Deployment ...

Kaito
1. Deploy Kaito on k8s 2. Deploy Workspace CRD 3. There's no step 3

```yaml
apiVersion: kaito.sh/v1alpha1
kind: Workspace
metadata:
    name: workspace-falcon-7b
resource:
    instanceType: "Standard_NC12s_v3"
    labelSelector:
        matchLabels:
            apps: falcon-7b
inference:
    preset:
        name: "falcon-7b"
```

KAITO workspace CR

Can KAITO customize open-source LMs to my data and use cases?

```yaml
apiVersion: kaito.sh/v1alpha1
kind: Workspace
metadata:
    name: workspace-tuning-phi-3-mini-pycoder
resource:
    instanceType: "Standard_NC6s_v3"
    labelSelector:
        matchLabels:
            apps: tuning-phi-3-mini-pycoder
tuning:
    preset:
        name: phi3mini128kinst
  method: qlora
  input:
     urls:
        - “…”
  output:
     image: …
```

KAITO fine tuning workspace CR

```
apiVersion: kaito.sh/v1alpha1
kind: Workspace
metadata:
     name: workspace-phi-3-mini-adapter
resource:
     instanceType: "Standard_NC6s_v3"
     labelSelector:
          matchLabels:
                apps: phi-3-adapter
inference:
     preset:
         name: phi3mini128kinst
       accessMode: private
       presetOptions:
             image: …
    adapters:
         - source:
                 name: “phi-3-coding-adapter”
                 image: …
            strength: “1.0”
```

KAITO inference with adapter workspace CR

KAITO roadmap

Inference model containerization and infra configuration/setup

Day 0

Day 1

Explore more LLMs and fine-tune/use RAG for your scenarios

Day 2

Train models with GPU sharing or batch process speed-up

Model Context Protocol with KAITO

Azure/**azure-mcp**

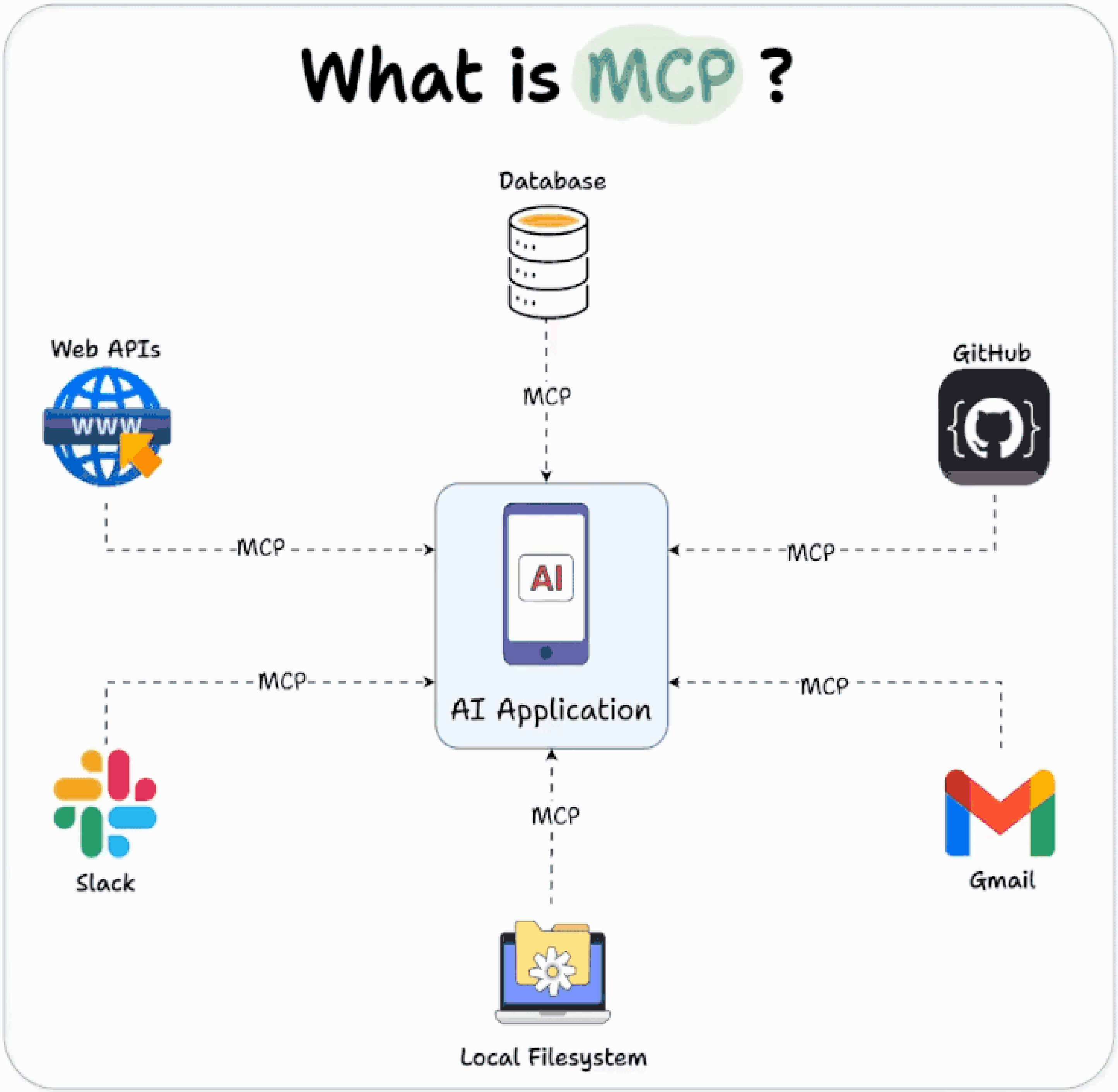
This repository is for development of the Azure MCP Server, bringing the power of Azure to your agents.

20 Contributors · 32 Issues · 489 Stars · 98 Forks

rohitg00/**awesome-devops-mcp-servers**

A curated list of awesome MCP servers focused on DevOps tools and capabilities.

1 Contributor · 0 Issues · 1 Star · 1 Fork



Summary

Key takeaways

1

AI has revolutionized the industry

2

Taking a cloud native approach makes adopting AI more flexible, streamlined, and cost-effective

3

Use AI to help you operate your cloud native infrastructure and apps

Join the KAITO community!

Get started with KAITO QuickStart on GitHub: aka.ms/kaito

Contribute a custom model seamlessly: aka.ms/kaito-models

Watch KAITO in action: aka.ms/kaito-live

Kubernetes AI Toolchain Operator (Kaito)

release v0.3.0 | go report A+ | Go v1.22.3 | codecov 61%

What is NEW!
Latest Release: July 12th, 2024. Kaito v0.3.0.
First Release: Nov 15th, 2023. Kaito v0.1.0.

Kaito is an operator that automates the AI/ML inference model deployment in a Kubernetes cluster. The target models are popular large open-sourced inference models such as falcon and llama2. Kaito has the following key differentiations compared to most of the mainstream model deployment methodologies built on top of virtual machine infrastructures:

- Manage large model files using container images. A http server is provided to perform inference calls using the model library.
- Avoid tuning deployment parameters to fit GPU hardware by providing preset configurations.
- Auto-provision GPU nodes based on model requirements.
- Host large model images in the public Microsoft Container Registry (MCR) if the license allows.

Using Kaito, the workflow of onboarding large AI inference models in Kubernetes is largely simplified.

Architecture

https://github.com/ams0/demos


THIS IS THE WAY .

Microsoft